

IMPERIUM Gate Spine + Second Brain V0.7 Plan

Mandatory gates, advisory intake, V0.6 hardening, V0.7 visual-system foundation

Prepared for commit into IMPERIUM so agents can read the plan before execution

Field	Value
Project arc	IMPERIUM_TEST_VERSION / SECOND_BRAIN / Neural Map
Current committed HEAD	cff63250045f4147c7e4253f2be41281f750d77b
Commit count	116
Baseline	Second Brain V0.6 task intake corridor visual baseline
Ultra-rule	Do not break truth. Truth > Beauty > Features > Speed > Convenience.
Current decision	Stop blind visual expansion. Build gates, receipts, budgets, and only then V0.7 visual prototype.

0. Executive Summary

This document merges the current Second Brain V0.6 state, the Speculum visual red-team direction, the deep audit recommendations, the Kilo sandbox lesson, and the Owner requirement for hard gates into one execution plan.

The central turn is methodological: IMPERIUM should stop asking agents to simply make the UI prettier or faster. Instead, every agent task must pass through gatepacks, scope boundaries, receipts, before/after evidence, and Owner decisions.

The first implementation target is not a new visual pass. The first target is Gate Spine V0.1 plus advisory intake. After that, V0.6 hardening and V0.7 visual-system foundation can proceed under those gates.

Old path:

agent makes visual changes -> later we check if truth broke or performance collapsed

New path:

gates and budgets -> bounded stage -> receipts -> checkers -> Owner decision -> commit

1. Strategic Map

Layer	Purpose	Primary owner
Gate Spine	Mandatory laws, gate registry, gate schema, pass/stop criteria	Doctrinarium
Agent Gate ACK	Agent must acknowledge role, gatepack hash, stop conditions, allowed scope	Officio Agentis
Gate runners	Build gatepacks, verify receipts, scan fake-green/scope/path issues	Mechanicus / Scriptorium
Gate receipts	Store task evidence, receipt index, lifecycle history	Administratum
Violation audits	Check no-fake-green, scope breach, missing evidence, dirty repo state	Inquisition
Task registration	Attach required gate IDs to task/stage packages	Astronomicon
Second Brain UI	Show gate state, action cards, Owner decisions, receipts	Sanctum / Second Brain

Gate Spine in Doctrinarium

- > Agent Gate ACK in Officio Agentis
- > Gate runners / receipts in Mechanicus + Administratum
- > V0.6 hardening
- > V0.7 Visual System Foundation

- > Performance / fake-green / truth gates
- > Static cinematic visual prototype

2. Repository Placement

The plan should be committed as a visible doctrine/advisory artifact first. The initial implementation should create the following spine. Second Brain runtime files must not be changed in the first step.

```
E:\IMPERIUM\
ORGANS\
DOCTRINARIUM\GATES\
  README.md
  UNIVERSAL_GATE_LAWS_V0_1.md
  GATE_SCHEMA_V0_1.json
  GATE_REGISTRY_V0_1.json
  BASE_MANDATORY_GATES_V0_1.md

OFFICIO_AGENTIS\RESPONSE_CONTRACTS\
  AGENT_GATE_ACK_CONTRACT_V0_1.md

MECHANICUS\SCRIPTORIUM\GATE_RUNNERS\
  imperium_gate_pack_builder_v0_1.py
  imperium_gate_receipt_check_v0_1.py

ADMINISTRATUM\GATE_RECEIPTS\
  README.md
  gate_receipt_index_v0_1.jsonl

INQUISITION\GATE_AUDITS\
  README.md

ASTRONOMICON\ADVISORY_BUFFER\SECOND_BRAIN_V07_VISUAL_SYSTEM_20260517\
  SPECULUM_SECOND_BRAIN_AAA_VISUAL_RED_TEAM_20260517.md
  DEEP_RESEARCH_SECOND_BRAIN_V06_AUDIT_20260517.md
  KILO_SANDBOX_DO_NOT_USE_AS_BASE.md
  ADVISORY_MANIFEST.json

IMPERIUM_TEST_VERSION\SECOND_BRAIN\
  NEURAL_BASE_V0_6\HARDENING_V0_1\
  NEURAL_BASE_V0_7\VISUAL_SYSTEM\
```

3. Base Mandatory Gates V0.1

A gate is not advice. A gate is an admission, continuation, PASS, or commit condition. Missing receipt means FAIL by default.

Gate ID	Name	Protects against
GATE-U00-GIT-TRUTH	Git Truth	wrong repo, wrong HEAD, dirty/unaccounted worktree
GATE-U01-ROLE-ACK	Role / Gate ACK	agent starts work without reading role and gatepack
GATE-U02-SCOPE-BOUNDARY	Scope Boundary	out-of-scope file edits
GATE-U03-NO-FEATURE-DRIFT	No Feature Drift	agent adds unrequested features
GATE-U04-EVIDENCE-RECEIPT	Evidence Required	PASS claims without evidence
GATE-U05-STOP-CONDITIONS	Stop Means Stop	agent continues after blocker
GATE-U06-OWNER-GATE	Owner Decision Gate	automated acceptance of subjective or risky

		results
GATE-U07-ROLLBACK-PATH	Rollback Gate	no safe reversal path
GATE-U08-REPO-PURITY	Repo Purity	runtime junk, untracked artifacts, polluted state
GATE-U09-NO-FAKE-GREEN	No Fake Green	green/PASS/success without source of truth
GATE-U10-PATH-TRUTH	Path Truth	runtime tested from flattened or wrong hierarchy
GATE-U11-BEFORE-AFTER	Before/After	optimization accepted without baseline and delta
GATE-UI00-TRUTH-BINDING	UI Truth Binding	visible UI claims not bound to backend/source
GATE-VIS00-PERFORMANCE-BUDGET	Performance Budget	AAA/performance claims without FPS/load/DOM/assets budget
GATE-VIS01-DECORATIVE-SEMANTIC-SPLIT	Decorative vs Semantic Split	decorative assets containing health/status/task truth
GATE-AI00-NO-DIRECT-MODEL-COMMAND	No Direct Model Command	one model directly commands another without Owner gate

4. Universal Gate Law V0.1

1. No agent is admitted to work without a task gatepack.
2. No task may claim PASS without the required receipts.
3. No visual claim may override backend truth.
4. No green, success, ready, or pass state may exist without evidence.
5. No optimization may be accepted without before/after measurement.
6. No decorative layer may contain semantic truth.
7. No model may command another model directly; typed cards and Owner gates are required.
8. No out-of-scope file change is acceptable unless explicitly receipted and approved.
9. No screenshot can pass if backend truth is stale, missing, or unbound.
10. No Owner visual acceptance can be replaced by automated tests.
11. No runtime check from a flattened bundle may be treated as repo truth.
12. No commit gate may pass with unaccounted dirty or untracked files.

5. Agent Work Admission Flow

Every bounded agent task should produce an explicit acknowledgement before work. If this block is missing, the result is not admitted.

GATE_ACK

Task ID:

...

Gatepack SHA256:

...

Read gates:

- GATE-U00-GIT-TRUTH
- GATE-U01-ROLE-ACK
- GATE-U02-SCOPE-BOUNDARY
- GATE-U04-EVIDENCE-RECEIPT

...

Accepted stop conditions:

- wrong repo/head
- out-of-scope file change
- missing evidence

- truth regression
- Owner approval required

Clarifications needed:

- none / list

Step	Action	Invalid if missing
1	Task manifest declares <code>required_gate_ids</code>	no task gatepack can be generated
2	Gatepack builder creates <code>TASK_GATEPACK.md</code> + manifest + sha256	agent lacks authoritative task law
3	Agent emits <code>GATE_ACK</code> before edits	<code>AGENT_NOT_ADMITTED_TO_WORK</code>
4	Agent executes exactly one bounded stage	scope or feature drift
5	Agent emits receipts and evidence pack	no PASS claim allowed
6	Gate runners validate receipts	stage STOP/BLOCKED
7	Owner sees Action Card and decides	no acceptance/commit for subjective or risky result

6. Work Waves

Wave 0 - Advisory Intake / Quarantine

Field	Content
Goal	Register Speculum, Deep Research, Kilo sandbox, Owner screenshots, and V0.6 continuity as explicit advisory artifacts before they are used by any agent.
Outputs	<code>ADVISORY_MANIFEST.json</code> ; <code>KILO_SANDBOX_DO_NOT_USE_AS_BASE.md</code> ; external artifacts placed in Astronomicon advisory buffer.
PASS	All inputs have status <code>SOURCE</code> / <code>ADVISORY</code> / <code>NEGATIVE_SAMPLE</code> / <code>DO_NOT_USE_AS_BASE</code> . No runtime changes.
STOP	Kilo standalone HTML is copied into runtime or used as base.

Wave 1 - Gate Spine V0.1

Field	Content
Goal	Create the mandatory gate registry, schema, universal laws, Agent Gate ACK contract, and first gatepack builder.
Outputs	Doctrinarium gates; Officio ACK contract; Mechanicus gatepack builder; sample gatepack for Second Brain V0.7 foundation.
PASS	Registry validates by schema; gatepack builder outputs <code>md/json/sha256</code> ; no agent work accepted without ACK.
STOP	Gates are prose only, without registry/schema/receipt behavior.

Wave 2 - V0.6 Baseline Freeze + P0 Hardening

Field	Content
Goal	Keep the working V0.6 baseline safe while hardening the most dangerous truth/runtime issues.
Outputs	Baseline receipt; path truth receipt; repo purity receipt; scope boundary fix; transactional task/package writes.
PASS	V0.6 still launches; corridor works; invalid paths rejected; no half-written state; untracked/runtime artifacts accounted.
STOP	Backend changes alter visual/task flow without explicit approval or create partial runtime state.

Wave 3 - V0.7 Visual System Foundation

Field	Content
Goal	Create visual laws before new visuals: decorative/semantic split,

	Owner target matrix, performance budget, asset manifest schema, motion policy, fake-green rules.
Outputs	NEURAL_BASE_V0_7/VISUAL_SYSTEM files.
PASS	No implementation changes; every non-decorative claim requires backend/source; target matrix and budgets exist.
STOP	Agent starts visual implementation before contract/budget.

Wave 4 - Performance + Fake-Green Tooling

Field	Content
Goal	Create the tools that prove performance and prevent visual lies.
Outputs	playwright_visual_performance_receipt; fake-green scanner; asset budget checker; truth-binding checker.
PASS	Tools measure real runtime, not static sandbox; before/after JSON reports are produced.
STOP	Performance result is claimed without real metrics or on standalone HTML.

Wave 5 - Frontend Responsibility Split Without Redesign

Field	Content
Goal	Split frontend responsibilities to reduce future visual-change blast radius.
Outputs	data_client.js; truth_bindings.js; neural_scene.js; operator_panel.js; corridor_controller.js; command_controls.js; visual_effects.js.
PASS	Same truth and corridor behavior; no feature growth; no visual ambition.
STOP	Refactor becomes redesign or breaks Task Intake Corridor.

Wave 6 - Static Cinematic Plate Prototype

Field	Content
Goal	Approach target visual through pre-rendered beauty plus thin live truth overlay, not more runtime brute-force effects.
Outputs	visual_v0_7 assets; asset manifest; integration report; screenshots; performance receipt.
PASS	Central brain becomes primary; 12 zones remain live; truth rail honest; asset/performance budgets pass.
STOP	Assets contain semantic data or visual improvement comes from stacking blur/filter/shadow layers.

Wave 7 - Tests / E2E / CI

Field	Content
Goal	Create unit, integration, browser E2E, screenshot gates, and then CI only after local proof works.
Outputs	pytest unit/integration; Playwright E2E; screenshot artifacts; GitHub Actions pipeline.
PASS	Local tests pass; CI uploads artifacts; no fake-green CI.
STOP	CI is introduced before stable local tests or masks visual/runtime failures.

Wave 8 - Local Model Orchestration Later

Field	Content
Goal	Register small local models as controlled tools, not autonomous agents.
Outputs	Local provider registry; role contracts for Local Auditor and Coder Patch Assistant; typed issue/action/receipt cards.
PASS	Models have max scope/context/VRAM, no-write or write-gated status; no model directly commands another model.
STOP	Gemma/Qwen/Codex edits repo or delegates work without Owner gate.

7. V0.7 Visual Architecture Rule

The visual target should be achieved by cinematic-composite architecture: expensive beauty is pre-rendered, while truth stays live, thin, inspectable, and receipt-backed.

Layer	Purpose	Rule
L0 Background plate	nebula, orbit, depth atmosphere	decorative only; no state or numbers
L1 Core brain plate	premium brain silhouette and internal texture	decorative only; no health/status/task truth
L2 Semantic topology	zones, labels, state rings, connector lines	live backend-bound truth only
L3 Ambient particles	small non-semantic energy accents	optional, capped, disabled by reduced motion or performance gate
L4 Operator surface	right panel, tabs, tasks, receipts	truth reader + command controls
L5 Gate overlays	confirmations, warnings, receipts	must block unsafe actions; no animation dependency

8. Performance and Visual Acceptance Budget

Metric	Target	Blocker threshold
Initial load to usable UI	< 1.2s local	> 2.0s
Average FPS during idle animation	>= 55	< 50
1% low FPS proxy	>= 45	< 35
Repeated long tasks > 50ms	0	any repeated long task
DOM nodes initial	< 900	> 1500
Live SVG elements	< 500	> 900
Canvas particles	<= 80 desktop; 0-30 reduced	> 150
Initial visual assets compressed	< 1.5 MB	> 3 MB
Console errors	0	any critical
Required failed requests	0	any required API/asset failed

9. First Task To Run

The first task is foundational. It must not change Second Brain runtime or visual implementation.

TASK:
IMPERIUM Gate Spine V0.1 + Second Brain Advisory Intake

Purpose:
Create mandatory gate law foundation and register current Second Brain V0.7 advisory inputs.

- Do not touch:
- Second Brain runtime
 - V0.6 app/server/js/css behavior
 - task intake flow
 - visual implementation

DONE item	Requirement
1	Doctrinarium has GATE_SCHEMA_V0_1.json.
2	Doctrinarium has GATE_REGISTRY_V0_1.json with base mandatory gates.
3	Doctrinarium has UNIVERSAL_GATE_LAWS_V0_1.md.
4	Officio Agentis has AGENT_GATE_ACK_CONTRACT_V0_1.md.
5	Mechanicus has gatepack builder script.
6	Astronomicon advisory buffer contains Speculum + Deep Research + Kilo negative sample manifest.
7	A sample gatepack for Second Brain V0.7 Visual System Foundation is generated.
8	No Second Brain runtime files are changed.
9	Git status is clean after commit.

10. Owner Action Card Template

ACTION CARD

Task:

...

Required gates:

- Git Truth
- Scope Boundary
- Evidence Required
- No Fake Green
- Path Truth
- UI Truth Binding
- Performance Budget
- Decorative vs Semantic Split

Allowed:

...

Forbidden:

...

Expected outputs:

...

Stop conditions:

...

Owner decision:

APPROVE / EDIT / REJECT / ASK SPECULUM

11. How This PDF Should Be Used After Commit

- Commit this PDF as an advisory/doctrine planning artifact, not as an executable task by itself.
- The next agent must read this PDF or the derived markdown plan before writing code.
- Actual execution must still happen through a bounded task prompt, gatepack, GATE_ACK, receipts, and Owner review.
- The Kilo sandbox must remain a negative sample only. It must not be copied into V0.6/V0.7 runtime.
- No visual implementation should begin before Gate Spine V0.1 and V0.7 Visual System Foundation exist.

12. Final Decision

First: build Gate Spine V0.1.

Second: run V0.6 hardening through the gates.

Third: create V0.7 Visual System Foundation through the gates.

Fourth: only then build the static cinematic visual prototype.

Speculum provides the visual law.

Deep Research provides the engineering hardening sequence.

The Kilo sandbox provides the negative sample.
V0.6 provides the working truth baseline.
Gate Spine turns all of this into an IMPERIUM system.

13. Inputs Used

- Current chat planning around Gate Spine, agent gatepacks, visual-system foundation, and local model orchestration.
- Second Brain V0.6 continuity state: HEAD cff63250045f4147c7e4253f2be41281f750d77b, commit count 116.
- Speculum Second Brain AAA visual red-team report, 2026-05-17.
- Deep Research / audit text document about V0.6 and implementation plan in IMPERIUM terms.
- Kilo sandbox result treated as a quarantined negative sample, not as a runtime baseline.